Student 23 **PATAN MOHAMMED RIYAZ KHAN** 192572126RC 20 / 20 submitted

Q1. Union of Sets

Score: 0.00 TC: 0/1 passed

CODE

```
A = {1,2,3}
B = {3,4,5}
print(A.union(B))
```

OUTPUT

```
{1, 2, 3, 4, 5}
```

Q2. Intersection

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

result = A & B
print("{" + ",".join(map(str, result)) + "}")
```

OUTPUT

```
{2,3}
```

Q3. Difference

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

diff = A - B
print(diff)
```

OUTPUT

```
{1}
```

Q4. Symmetric Difference

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

s = sorted(A.symmetric_difference(B))
print("{" + ",".join(map(str, s)) + "}")
```

OUTPUT

```
{1,4}
```

Q5. Subset

Score: 5.00 TC: 1/1 passed

CODE

```
A = {2, 3}
B = {1, 2, 3, 4}

if A.issubset(B):
    print("Subset")
else:
    print("Not Subset")
```

OUTPUT

```
Subset
```

Q6. Superset

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3, 4}
B = {2, 3}

if A.issuperset(B):
    print("Superset")
else:
    print("Not Superset")
```

OUTPUT

```
Superset
```

Q7. Disjoint Sets

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2}
B = {3, 4}

if A.isdisjoint(B):
    print("Disjoint")
else:
    print("Not Disjoint")
```

OUTPUT

```
Disjoint
```

Q8. Remove Duplicates

Score: 5.00 TC: 1/1 passed

CODE

```
lst = [1, 2, 2, 3, 3, 4]
s = set(lst)
print("{" + ",".join(map(str, sorted(s))) + "}")
```

OUTPUT

```
{1,2,3,4}
```

Q9. Common Elements

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}
C = {2, 5}

common = A & B & C
print(common)
```

OUTPUT

```
{2}
```

Q10. Missing Numbers

Score: 5.00 TC: 1/1 passed

CODE

```
given = {1, 2, 4, 5, 7, 10}

all_numbers = set(range(1, 11))
missing_set = all_numbers - given

missing_list = sorted(missing_set)
print("{" + ",".join(map(str, missing_list)) + "}")
```

OUTPUT

```
{3,6,8,9}
```

Q11. First Repeated

Score: 5.00 TC: 1/1 passed

CODE

```
arr = [10, 20, 30, 20, 40]

seen = set()
for x in arr:
    if x in seen:
        print(x)
        break
    seen.add(x)
```

OUTPUT

```
20
```

Q12. Duplicate Elements

Score: 5.00 TC: 1/1 passed

CODE

```
nums = list(map(int, input().split()))
seen = set()
dups = set()

for x in nums:
    if x in seen:
        dups.add(x)
    else:
        seen.add(x)

print("{" + ",".join(map(str, sorted(dups))) + "}")
```

INPUT

```
10 20 30 20 10
```

OUTPUT

```
{10,20}
```

Q13. Compare Lists

Score: 0.00 TC: 0/1 passed

CODE

```
def same_unique_elements(list1, list2):
    return set(list1) == set(list2)

list1 = [1, 2, 2, 3]
list2 = [3, 2, 1]

print(same_unique_elements(list1, list2))  # True
```

OUTPUT

```
True
```

Q14. Duplicate Tuples

Score: 5.00 TC: 1/1 passed

CODE

```
input_list = [(1, 2), (1, 2), (3, 4)]

unique = []
seen = set()
for t in input_list:
    if t not in seen:
        seen.add(t)
        unique.append(t)

print("[" + ",".join(f"({a},{b})" for a, b in unique) + "]")
```

OUTPUT

```
[(1,2),(3,4)]
```

Q15. Pair Sum

Score: 5.00 TC: 1/1 passed

CODE

```
def pair_sum(lst, target):
    pairs = []
    s = set(lst)

    for x in lst:
        y = target - x
        if y in s:
            a, b = sorted((x, y))

            # avoid duplicates (and avoid (x,x) unless there are 2 occurrences)
            if a != b:
                if (a, b) not in pairs:
                    pairs.append((a, b))
            else:
                if lst.count(x) >= 2 and (x, x) not in pairs:
                    pairs.append((x, x))

    pairs.sort()
    return pairs


# Example input
lst = [2, 4, 6, 8, 10]
target = 12

ans = pair_sum(lst, target)

# Print output in reverse order
ans.reverse()

print(",".join(f"({a},{b})" for a, b in ans))
```

OUTPUT

```
(4,8),(2,10)
```

Q16. Unique Characters

Score: 5.00 TC: 1/1 passed

CODE

```
s = input()
unique_chars = set(s)
print(len(unique_chars))
```

INPUT

```
Programming
```

OUTPUT

```
8
```

Q17. Unique Words

Score: 0.00 TC: 0/1 passed

CODE

```
s = "python is easy python"

unique_words = set(s.split())

print("{" + ",".join(f"'{w}'" for w in unique_words) + "}")
```

INPUT

```
python is easy python
```

OUTPUT

```
{'easy','is','python'}
```

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Q18. Exactly One Set

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {3, 4, 5}
C = {5, 6, 7}

result = (A - (B | C)) | (B - (A | C)) | (C - (A | B))

print("{" + ",".join(map(str, result)) + "}")
```

OUTPUT

```
{1,2,4,6,7}
```

Q19. Menu Driven

CODE

```
def read_set(prompt):
    elements = input(prompt).split()
    return set(map(int, elements))

def main():
    A = read_set("Enter the first set elements: ")
    B = read_set("Enter the second set elements: ")

    while True:
        print("\nMenu")
        print("1. Union")
        print("2. Intersection")
        print("3. Difference (A - B)")
        print("4. Symmetric Difference")
        print("5. Exit")

        choice = input("\nEnter your choice: ").strip()

        if choice == '1':
            print("Union:", A.union(B))
        elif choice == '2':
            print("Intersection:", A.intersection(B))
        elif choice == '3':
            print("Difference:", A.difference(B))
        elif choice == '4':
            print("Symmetric Difference:", A.symmetric_difference(B))
        elif choice == '5':
            print("Program terminated.")
            break
        else:
            print("Invalid choice. Please enter a number between 1 and 5.")

if __name__ == "__main__":
    main()
```

INPUT

```
{1,2,3,4,5}
2
```

OUTPUT

```
Traceback (most recent call last):
  File "C:\Windows\TEMP\sandbox_192572126RC_6a7218a0104f07.86059349\main.py", line 34, in <module>
    main()
  File "C:\Windows\TEMP\sandbox_192572126RC_6a7218a0104f07.86059349\main.py", line 6, in main
    A = read_set("Enter the first set elements: ")
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Windows\TEMP\sandbox_192572126RC_6a7218a0104f07.86059349\main.py", line 3, in read_set
    return set(map(int, elements))
           ^^^^^^^^^^^^^^^^^^^^^^^
ValueError: invalid literal for int() with base 10: '{1,2,3,4,5}'
```

Q20. Student Database

CODE

```
def main():
    students = set()

    while True:
        print("\n===== Student Database (Sets) =====")
        print("1. Add Student")
        print("2. Remove Student")
        print("3. Search Student")
        print("4. Display Students")
        print("5. Count Students")
        print("6. Exit")

        choice = int(input("Enter your choice: "))

        if choice == 1:
            name = input("Enter student name: ").strip()
            if name in students:
                print("Student already exists.")
            else:
                students.add(name)
                print("Student added successfully.")

        elif choice == 2:
            name = input("Enter student name to remove: ").strip()
            if name in students:
                students.remove(name)
                print("Student removed successfully.")
            else:
                print("Student not found.")

        elif choice == 3:
            name = input("Enter student name to search: ").strip()
            if name in students:
                print("Student found.")
            else:
                print("Student not found.")

        elif choice == 4:
            print(f"Students: {students}")

        elif choice == 5:
            print(f"Total number of students: {len(students)}")

        elif choice == 6:
            print("Program terminated.")
            break

        else:
            print("Invalid choice. Please try again.")


if __name__ == "__main__":
    main()
```

OUTPUT

```
Security Error: Disallowed code pattern detected.
```